Compositions of Transformations and Symmetry

Combining translations, symmetry of regular polygons, reflections in pairs, and naming the single equivalent motion

- Apply the transformations in the order they are written — the first one acts on the original point, the second on that image.
- A regular polygon with n sides has n lines of symmetry, and rotating it by $360°/n$ about its centre maps it exactly onto itself.
- Reflecting twice in parallel lines gives a translation; reflecting twice in intersecting lines gives a rotation about the crossing point.

1. A figure is translated by $(x, y) \to (x + 3,\ y - 5)$ and the image is then translated by $(x, y) \to (x - 8,\ y + 2)$. A single translation has the same effect.

Horizontal component _________ Vertical component _________

2. A regular decagon is rotated about its centre. Find the smallest positive angle of rotation that maps the decagon exactly onto itself.

Answer: _________________

3. Point $P(4, -6)$ is reflected in the x-axis, and that image is then reflected in the y-axis, giving P''.

(a) Write the coordinates of P''. P'' (________, ________)

(b) One single transformation has the same effect as this pair. Name it, giving its centre and its angle, and say what in your answer to (a) shows it.

4. A figure is reflected in the vertical line $x = 1$, and that image is reflected in the vertical line $x = 5$. The result is a translation. Find the horizontal distance that translation moves every point.

Answer: ________________

5. Point $A(5, -1)$ is translated by $(x, y) \to (x - 2,\ y + 7)$, and the image is then given a half turn about the origin, giving A''.

A'' (________, ________)

6. The smallest rotation that maps a certain regular polygon onto itself is $24°$. Find the number of sides the polygon has.

Answer: ________________

7. A figure is reflected in one line through the origin and then in a second line through the origin. The two lines meet at an angle of $35°$. The result is a rotation about the origin. Find the angle of that rotation.

Answer: ________________

8. Point $Q(-2, 5)$ is turned a quarter turn counterclockwise about the origin, and that image is then reflected in the line $y = x$. The final image keeps the x-coordinate -2.

(a) Find the y-coordinate of the final image. ________

(b) One single transformation has the same effect as this pair. Name it, and say what in your answer to (a) shows it.

9. A regular hexagon is rotated about its centre.

(a) Find the smallest positive angle that maps it onto itself. __________

(b) Two of those smallest rotations are performed one after the other. Find the angle of the single rotation that results. __________

(c) Explain why performing six of those smallest rotations returns every vertex to where it started.

10. Segment $\overline{RS}$ has endpoints $R(1, 2)$ and $S(6, 2)$. It is reflected in the line $y = x$, and that image is then translated by $(x, y) \to (x + 3,\ y - 4)$. The final image of S is $S''(5, 2)$.

(a) Write the coordinates of R''. R'' (__________, __________)

(b) Find the length RS. __________

(c) Find the length $R''S''$. __________

(d) Explain why the two lengths had to come out the same.